

Internet of Things

By

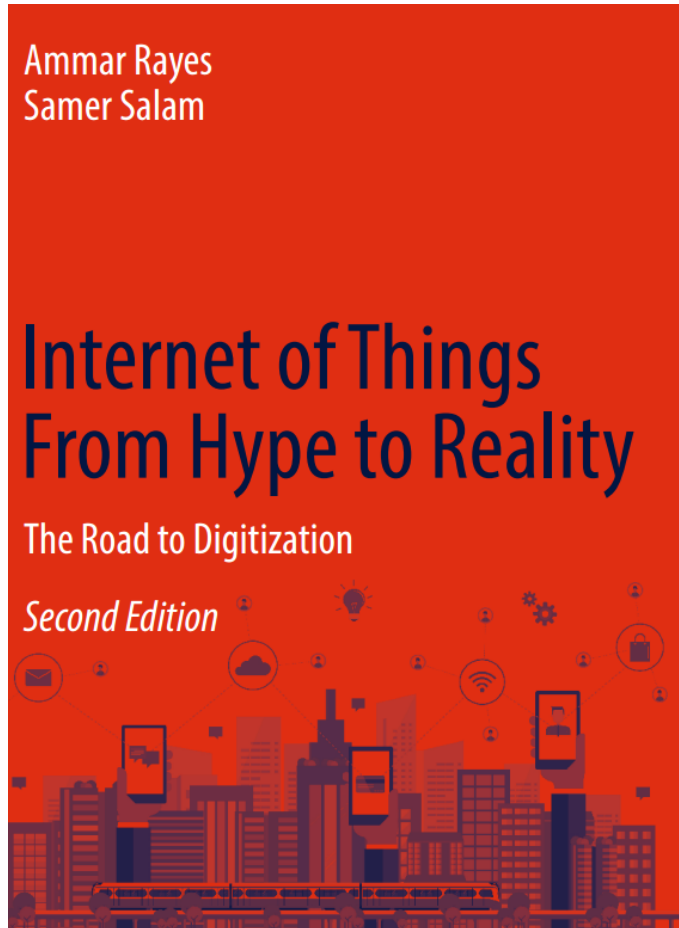
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Reference book



Chapter 6

Outline

- ▶ Defining Fog Computing
- ▶ Drivers for Fog
 - ▶ Data Deluge
 - ▶ Rapid Mobility
 - ▶ Reliable Control
 - ▶ Data Management and Analytics
- ▶ Characteristics of Fog
- ▶ Enabling Technologies and Prerequisites
 - ▶ Virtualization Technologies
 - ▶ Network Support for Mobility
 - ▶ Fog Orchestration
 - ▶ Data Management

Defining Fog Computing

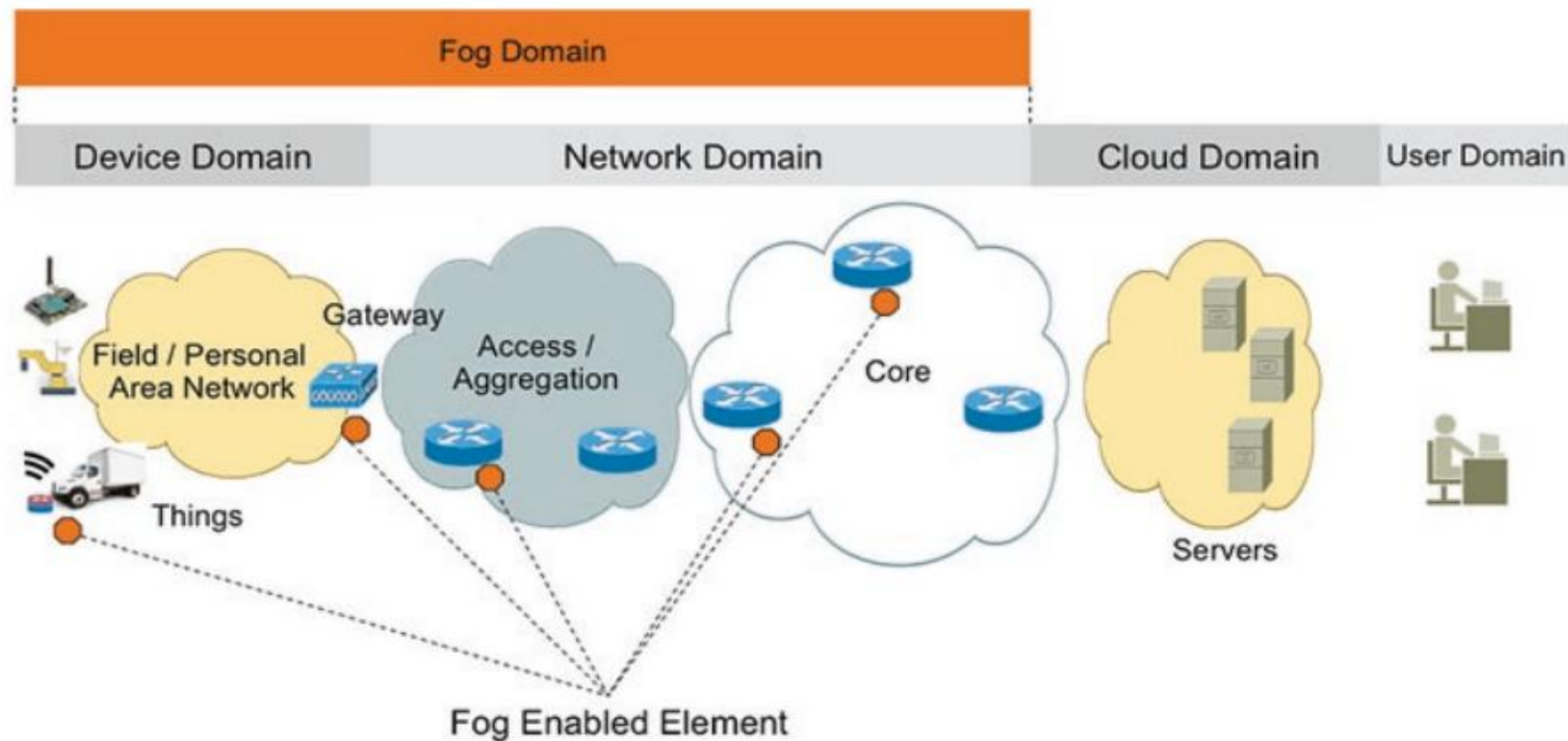


Fig. 6.1 Fog and Cloud

Defining Fog Computing

- ▶ Difference between cloud computing and fog computing
- ▶ Cloud computing refers to a model that provides users with on-demand access of a shared pool of computing resources over a network. These resources can be quickly provisioned and released through a self-service model. One of the key characteristics of the Cloud computing model is the notion of resource pooling, where workloads associated with multiple users (or tenants) are typically collocated on the same set of physical resources.
- ▶ Fog computing, refers to a platform for integrated compute, storage and network services that are highly distributed and virtualized. This platform can extend in locality from IoT end devices and gateways all the way to Cloud data centers but is typically located at the network edge. Fog augments Cloud computing and brings its functions closer to where data is produced (e.g., sensors) or needs to be consumed (e.g., actuators). Fog is not an alternative to Cloud computing;

Defining Fog Computing

- ▶ Fog computing(FC): “A decentralized Computing infrastructure based on Fog Computing nodes (FCNs) placed at **any point** of the architecture between the end devices and the cloud. The FCNs are **heterogeneous in nature** and thus can be based on different kinds of elements including but not limited to routers, switches, access points, IoT gateways as well as set-top boxes.
- ▶ Fog nodes gets the data feed from the sensors in real time.
- ▶ Response time is very minimal ideally in milliseconds.

Characteristics

Type of Implementation	FC	CC
Location	Near end device, dense and distributed	Local/Outdoor Installation in one place
Device	Routers, Switches, Access points, gateways...	Compact-size data centers
Access Mediums(mostly)	WiFi, LTE, ZigBee, MQTT, Bluetooth...	WiFi...
Logical Proximity	One/multiple hops	One hop
Ability for near-real-time Interaction	High	Medium
Multi-tenancy	Supported	Supported
Computation power	Medium	High

Application and use cases

- Smart Cities
- RAN-aware Context Optimization
- Augmented Reality
- E-Health
- Autonomous Vehicles
- Smart Grid
- Video Caching & Analysis

Due to features: Latency, Extensibility, Context Awareness, Power Consumption, Scalability, Privacy & Security

Fog computing

- ▶ Data Deluge(flood)
- ▶ The billions of devices that are projected to be connected to the Internet will create the data deluge problem.

Moore's Law stipulates that compute and storage technologies will double in capacity every 18 months.

Nielsen's Law, on the other hand, projects that the bandwidth at the network edge doubles every 24 months.

Cloud where will ultimately need to be processed and potentially stored. This allows the network to perform processing, analysis, and storage of data in lieu of blindly pushing all data up to the Cloud. With that, Cloud computing is brought closer to the data sources, the Things, which gives rise to the notion of Fog computing. Cloud becomes Fog when it is closer to Things, pun intended.

Fog computing

- ▶ **Rapid Mobility**
- ▶ Certain IoT use cases require support for rapid mobility of Things, for example, sensors on a **speeding vehicle communicating with roadside infrastructure** or a **passenger commuting on a train**. Due to rapid mobility, network conditions may vary frequently, due to signal fading, interference, or other conditions.
- ▶ **Multiple variables** will typically be at play to contribute to these characteristics, including **radio coverage, interference**, and the amount of resources shared with other mobile nodes.

Reliable Control

- ▶ IoT applications that focus on closed-loop control and actuation often required to produce the control decision have intensive computational and considerable storage demands.
- ▶ The combination of the above characteristics makes it unpalatable to rely on Cloud computing to support reliable real-time control with fixed latency. This is where Fog computing can complement the Cloud to address that IoT application niche.

Data Management and Analytics

- ▶ In terms of the number of devices generating data, widespread geographic footprint where these devices are deployed, vast amounts of data that need to be collected, aggregated, processed, and exposed to consuming entities, as well as real-time analytics or closed-loop control.
- ▶ For such class of applications, a data management and analytics platform that can handle the scale and performance requirements is needed.
- ▶ For data management and analytics, this operating paradigm is even more relevant because the IoT data often needs to be operated on within a context, which is well known at the edge of the network, close to the data sources, and is often lost or is irrelevant as the data travels deeper in the network and into the Cloud.

Characteristics of Fog

- ▶ The Fog and the Cloud both comprise of the same three building blocks: **compute**, **storage**, and **networking**.

First are the network edge location, location awareness, and low latency. Fog locates the services close to the data sources and consumers where it is possible to enrich the data with location context and operate on it with minimal latency.

Second is geographical and architectural distribution. This is in stark contrast to the Cloud model where all services are centralized in the data center.

Third is the extremely large number of nodes. While the Cloud drives demand for massively scalable data centers (MSDC), the Fog pushes the envelope further on scalability.

Characteristics of Fog

- ▶ **Fourth is mobility of nodes and endpoints.** The data sources, consumers, compute, or storage resources can all be mobile.
- ▶ **Fifth is real-time interaction.** In the Fog, the focus is on real-time analysis of streaming data as opposed to batch processing. Fog requires analysis of Data in Motion as opposed to Data at Rest.
- ▶ **Sixth is predominance of wireless access.** In the Cloud, connectivity relies on wireline technologies. The Fog will be mostly connected over wireless links, both because of the impracticality of running wires everywhere and to support the mobility requirements technologies, predominantly Gigabit Ethernet.
- ▶ **Seventh is the heterogeneity of resources.** In the Cloud, a given data center is managed by a single business entity, which goes about deploying homogeneous resources in order to minimize complexity and operational costs. With the Fog, the architecture is federated over resources managed by different business entities.

Enabling Technologies and Prerequisites

- ▶ The realization of the vision of Fog computing relies on a number of technologies that provide enabling building blocks and are key prerequisites for the architecture. These include lightweight compute virtualization, network mobility, orchestration, and application enablement technologies

Table 6.1 Summary comparison of Cloud and Fog computing

Requirement	Cloud computing	Fog computing
Latency and jitter	High/medium	Low
Location of service	Within Internet	Network edge
Distance between data sources/ consumers	Multiple hops	Single hop
Location awareness	No	Yes
Geo-distribution	Centralized (data center)	Distributed
Number of nodes	Large	Larger
Support for mobility	No	Yes
Data analytics	Data at Rest	Data in Motion
Connectivity	Wire line	Wireless

Virtualization Technologies

- ▶ Inherent to Fog computing is the ability to locate compute functions close to data producers and/or consumers. This assumes the availability of lightweight compute **virtualization technologies that allow workloads to be instantiated, as needed, on Fog nodes**. The latter act as shared compute resources among potentially a multitude of IoT applications.
- ▶ Virtualization technologies combine computing resources to present one or more operating environments using techniques such as hardware and software partitioning or aggregation, hardware emulation, resource sharing or time multiplexing.

Virtualization Technologies

- ▶ Virtualization advantages:
- ▶ It enables consolidation of both hardware and applications, thereby eliminating the expense associated with procuring and managing underutilized infrastructure.
- ▶ It also enables sandboxing, i.e., providing application with secure isolated execution environments. Virtualization also provides the flexibility of supporting multiple simultaneous operating systems over the same hardware infrastructure.
- ▶ It eases the migration of software stacks and allows the packaging of applications as stand-alone appliances.
- ▶ virtualization enables the portability and mobility of applications from one hardware or physical location to another with ease

Virtualization Technologies

- Virtualization at the CPU instruction set level allows an “emulator” to provide to an application the illusion of running on one processor architecture, whereas the real hardware actually belongs to a different architecture.
- Virtualization at the hardware abstraction layer level involves a virtual machine manager, or hypervisor, is a software layer that sits above the physical hardware and provides a virtualized view of all its services.
- Virtualization at the operating system level relies on virtualization software that runs on top of or as a module within the operating system.

VM and Containers

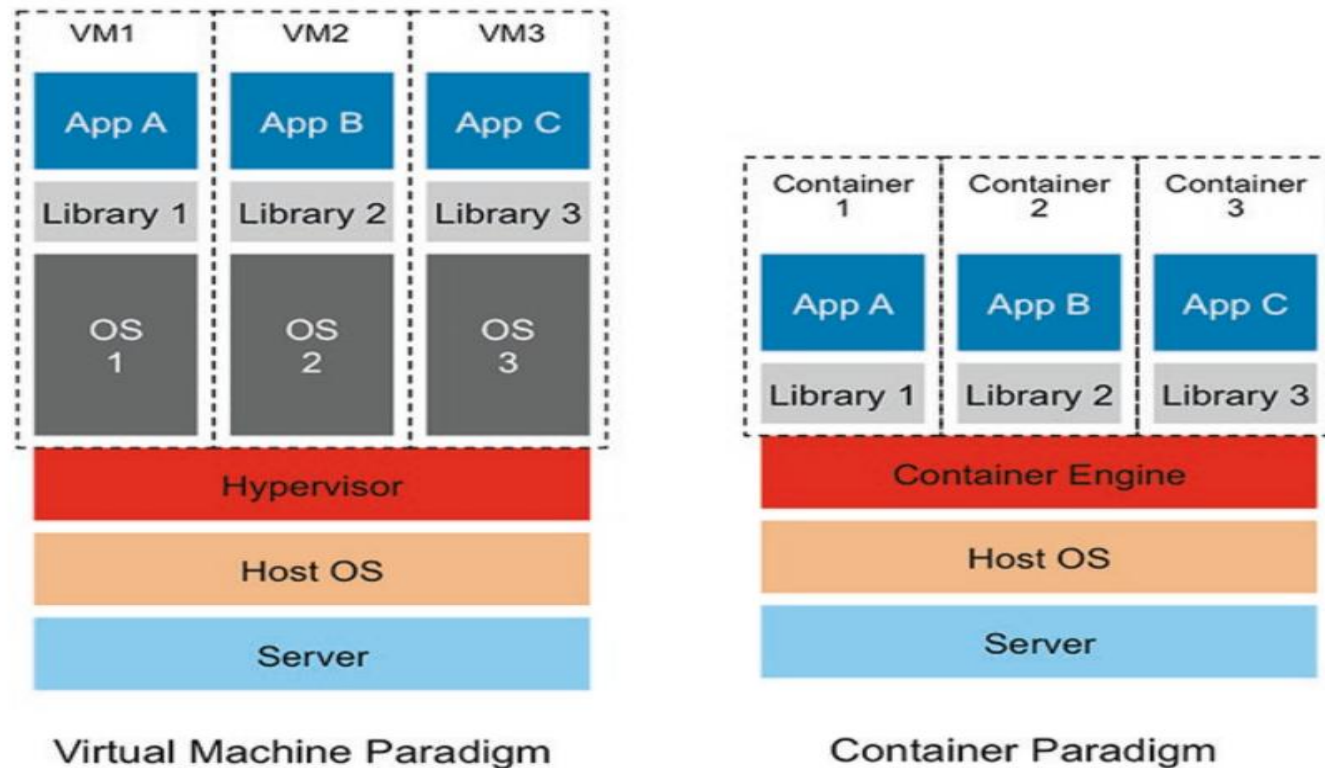


Fig. 6.2 VMs and containers

Both containers and virtual machines are popular virtualization constructs employed in Cloud computing today. Each of the two technologies has its own set of advantages and trade-offs. Virtual machines (VMs) are a virtualization technology at the hardware abstraction layer level.

VM and Containers

- ▶ They include portions of the operating system and select libraries: the minimal pieces that are absolutely required to run the application. Containers share the same operating system and, where applicable, common libraries.
- ▶ Due to this, containers are lighter-weight when compared to VMs, both in terms of their memory as well as processing requirements. As a result, given a specific hardware (e.g., a server) with a fixed resource profile, it is possible to support more containers than VMs running concurrently.
- ▶ This gives containers a clear scalability advantage over VMs, not only for Cloud computing but also for the Fog.
- ▶ Fog context: they are faster to migrate from one hosting node to another, a matter which characterizes them with the nimbleness required to support rapid mobility

VM and Containers

- ▶ A set of trade-offs:
- ▶ since containers share the same underlying operating system, it is not possible to use them to deploy applications that require disparate operating system environments, or different OS versions, on the same physical hardware. (Such restriction does not apply to virtual machines, since they include their own copy of the operating system.)
- ▶ In a container to be subjected to security threats due to malicious or misbehaving code running in another container on the same operating system.
- ▶ With virtual machines, the security threat is smaller in comparison, because the attack surface is minimized due to the fact that each VM has an independent operating system instance.
- ▶ Linux, the leading open operating system platform, supports both virtual machines and containers.
- ▶ Containers and VMs both provide the capability to sandbox Fog applications from one another and to control their resource usage.

Docker

- ▶ Docker is an open source project that provides a packaging framework to simplify the portability and automate the deployment of applications in containers. Docker introduces scripts composed of a series of instructions that automate the deployment process from start to finish. These scripts are referred to as “Dockerfiles”.
- ▶ Docker defines a format for packaging an application and all its dependencies into a single portable object. The portability is guaranteed by providing the application a runtime environment that behaves exactly the same on all Docker-enabled machines. Docker also provides tooling for container version tracking and management.



Application Mobility

- ▶ Virtualization technologies decouple the application software from the underlying compute, storage, and networking resources. As such, it enables unrestricted workload placement and mobility across geographically dispersed physical resources.
- ▶ In the latter, a VM that is powered on and operational is moved across hosts, without any interruption to its operation.
- ▶ In order to ensure seamless mobility in the case of “live” migration, the VM retains its original Internet Protocol (IP) and Medium Access Control (MAC) addresses.

Network Support for Mobility

- ▶ Rapid mobility is one of the drivers for Fog computing. **Fog deployment must support seamless mobility of the communicating endpoints.**
- ▶ Networking systems rely on the address of the endpoints in order to deliver messages to their intended recipients. Depending on the technology at hand, the address either connotes the identity or the location of the endpoint. For example, Media Access Control (MAC) addresses are identity addresses, because they are burnt into the machine and uniquely identify it on a network.
- ▶ Applications that are deployed in a virtualization construct, such as a virtual machine, can perform seamless mobility. **With seamless mobility, the application's MAC and IP addresses remain unchanged as the associated VM moves from one physical server node to another.**

Network Support for Mobility

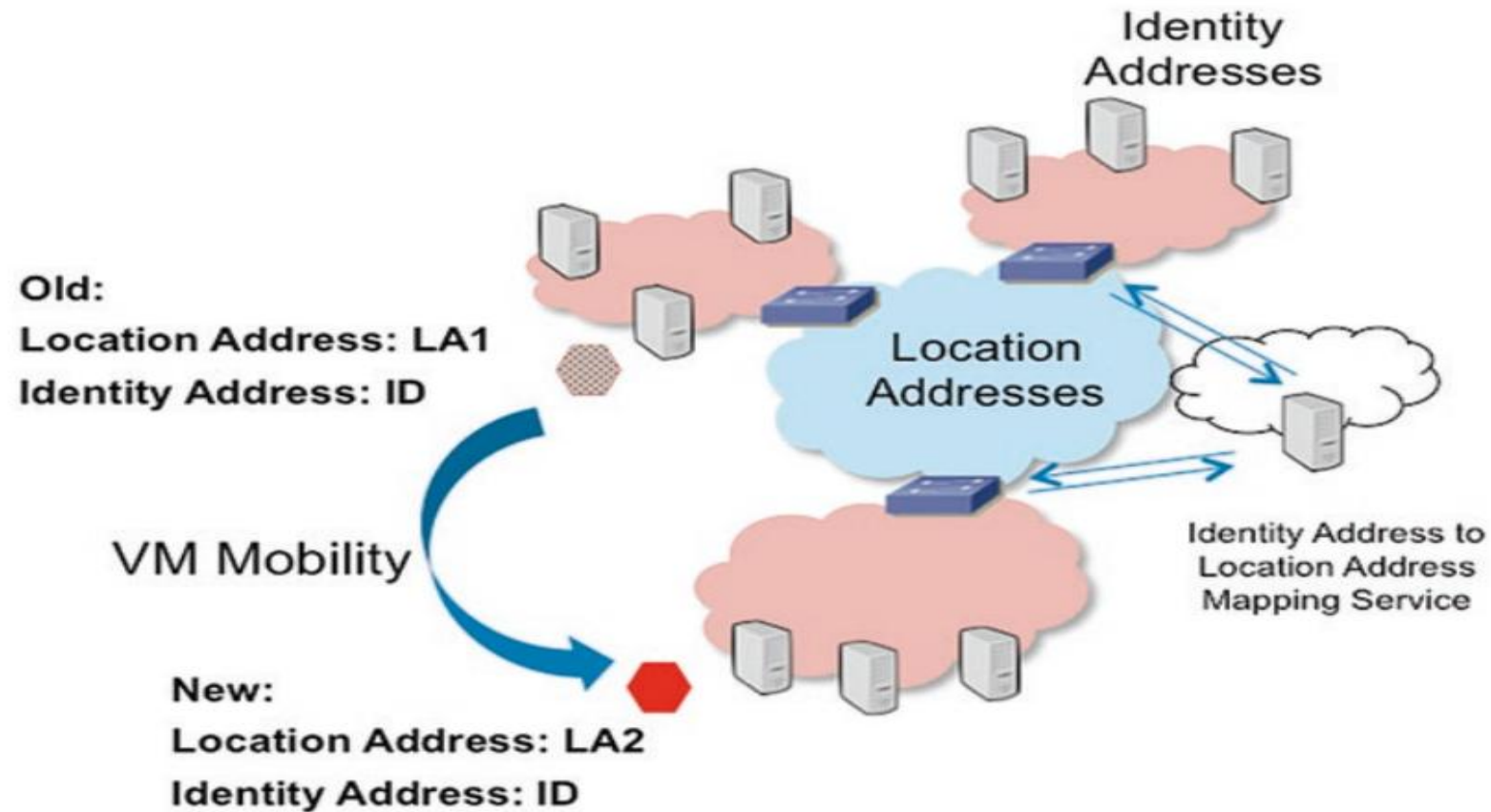


Fig. 6.3 Identity vs. location addresses with application mobility

Ethernet Virtual Private Network

- ▶ Ethernet Virtual Private Network (EVPN) is an overlay technology that allows Layer 2, and even Layer 3, virtual private networks to be created over a shared Internet Protocol (IP) or Multiprotocol Label Switching (MPLS) transport network.

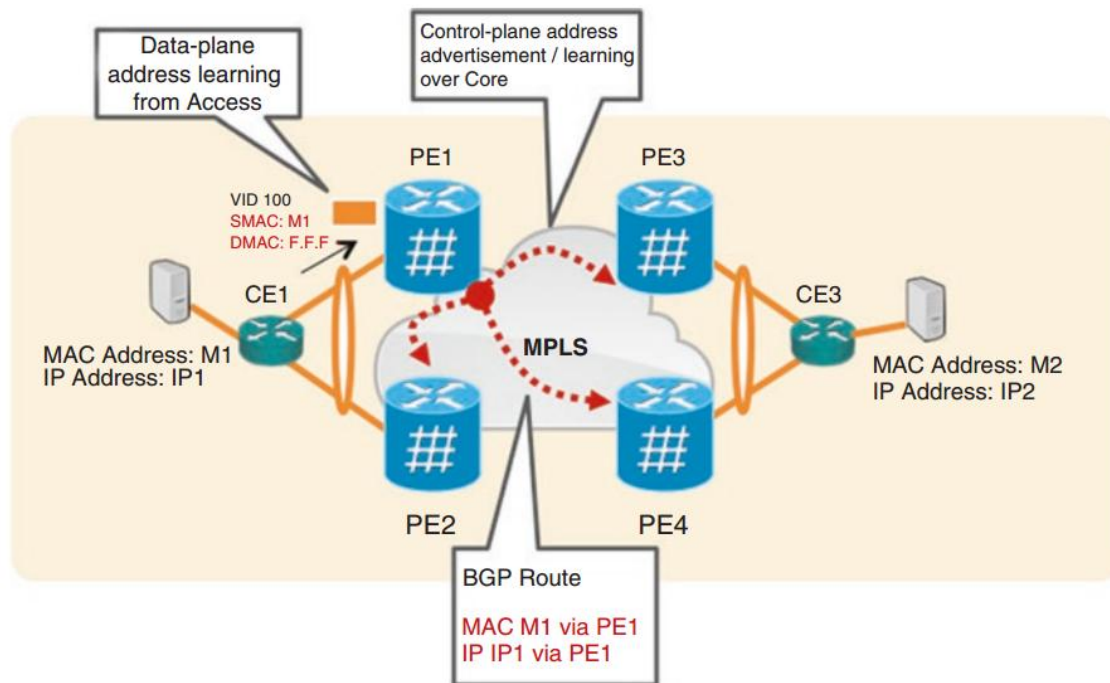
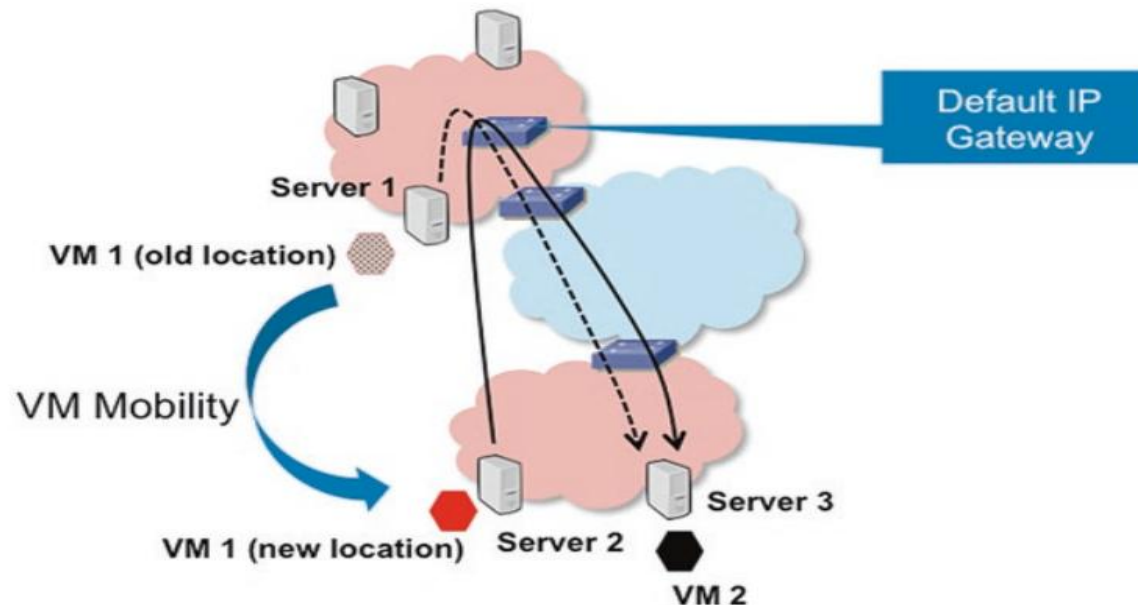


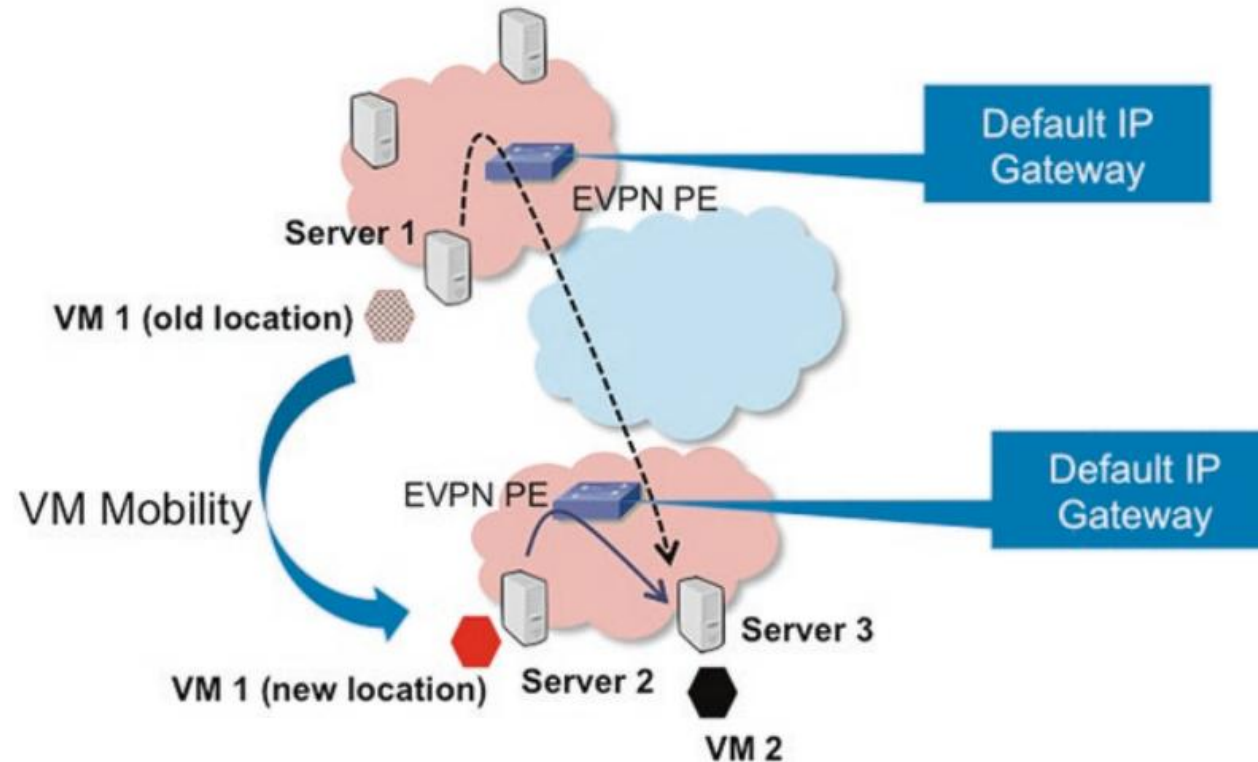
Fig. 6.4 Ethernet Virtual Private Network (EVPN) architecture

Default IP Gateway Problem

- ▶ As a VM moves from one physical server to another, both its memory (RAM) and disk image are maintained unchanged. This means that the VM's configuration remains unmodified. The configuration includes, among other things, the address of the Default IP Gateway that the VM should use in order to forward network traffic to remote nodes.



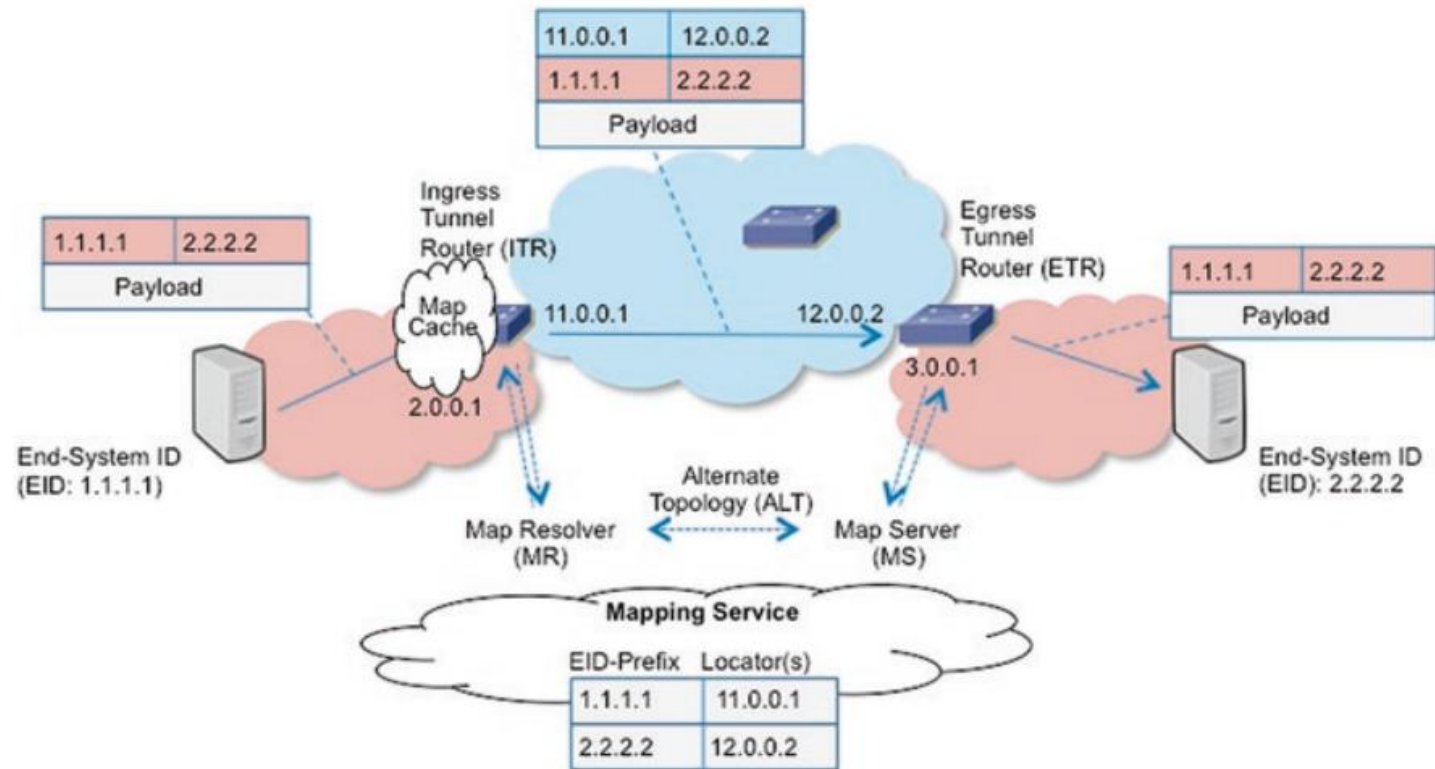
Locator/Identifier Separation Protocol (LISP)



Locator/Identifier Separation Protocol (LISP) is an overlay networking solution that allows complete decoupling of the addressing structure of end-hosts from that of the network infrastructure. LISP formally defines two namespaces for IP addresses

Fig. 6.7 EVPN Default Gateway solution

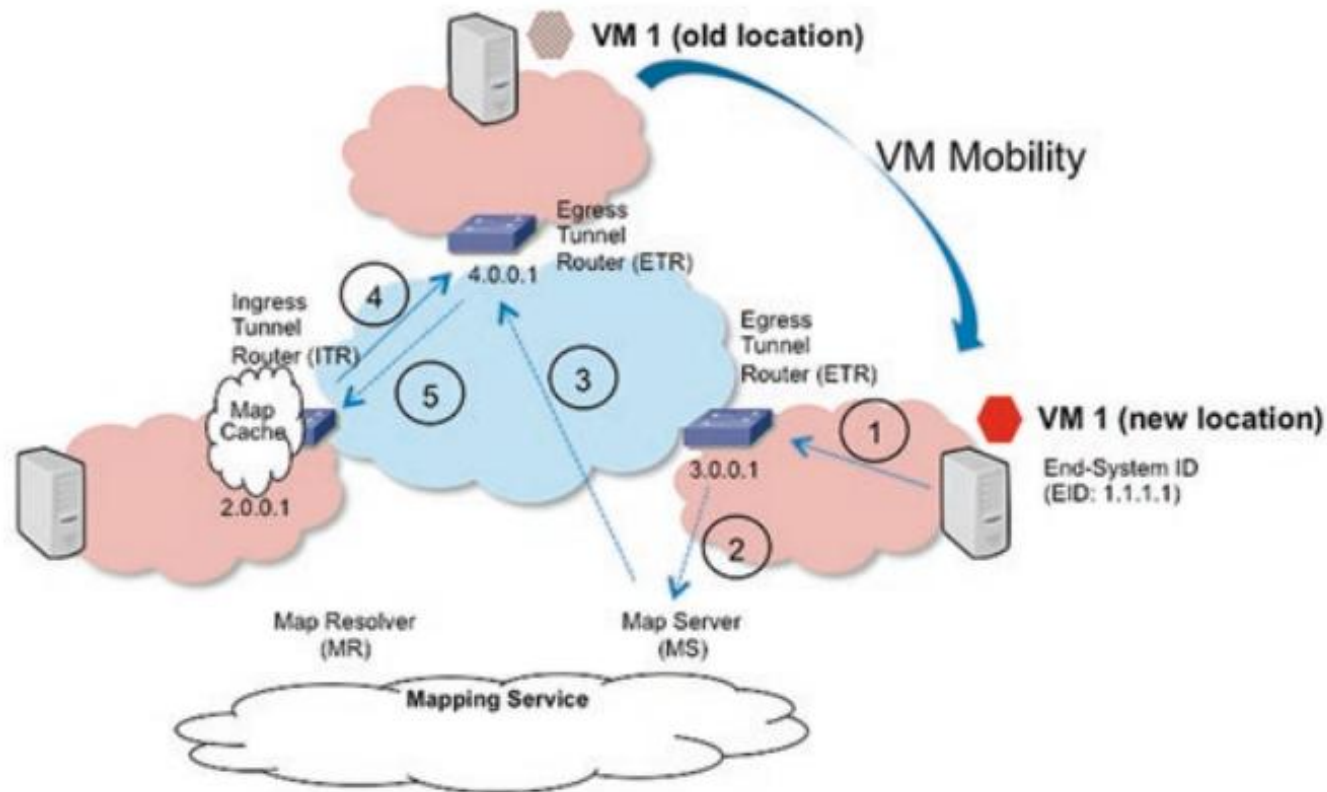
LISP architecture



LISP architecture

LISP mobility

1. VM sends traffic from new location to new ETR
2. ETR updates Map Server with new EID/ ETR mapping
3. Map Server deregisters EID from old ETR
4. ITR sends traffic to EID to old ETR
5. Old ETR sends Solicit-Map-Request to ITR to update its Map Cache



LISP mobility

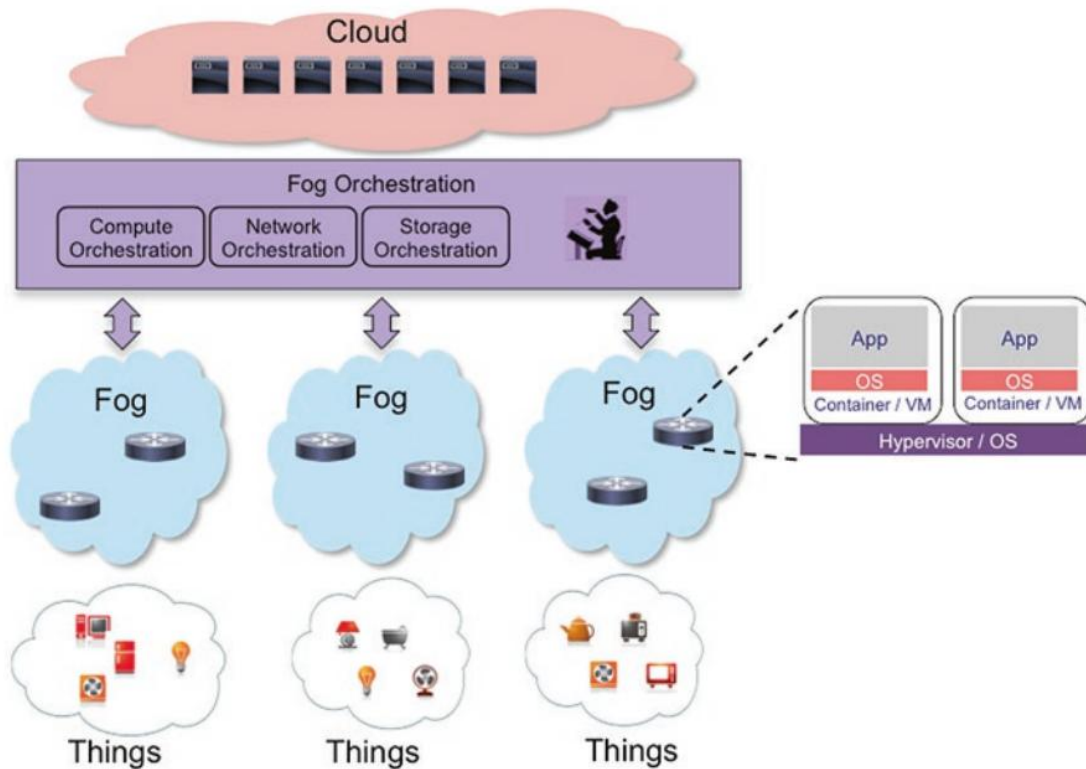
Default IP Gateway Problem

- ▶ LISP solves the Default IP Gateway Problem by ensuring that every site has a default gateway configured for the same prefix. This gateway must use the same (virtual) IP and MAC Addresses in order to guarantee that the traffic originating from the moved VM follows an optimal path out of the local LISP Tunneling Router rather than being forwarded to another site.
- ▶ First Hop Redundancy Protocols (e.g., VRRP) must be configured with identical gateway and MAC addresses in all sites, and their packets must not be allowed to leak beyond a given site. This way, when a VM moves it will always find the same default gateway regardless of its location

Fog Orchestration

- ▶ Orchestration, in the context of Fog computing, refers to the process of automating the various workflows that perform the full lifecycle management of the Fog infrastructure.
- ▶ This includes the provisioning and management of its three components (compute, network, storage) and associated resources.
- ▶ For illustration, tasks such as deploying, debugging, patching, and updating applications or operating systems, setting up network connectivity between application entities and reserving bandwidth, and allocating and expanding disk space are all examples of workflows that fall under orchestration.

Fog Orchestration



Fog orchestration differs from Cloud orchestration in three different facets:
topology, Things connectivity, and network performance guarantees.

Fig. 6.10 Fog orchestration

Things Connectivity

- ▶ With Fog, the orchestration software needs to be able to deploy applications, which need direct access to Things (e.g., legacy applications), on Fog nodes that are physically connected to these specific Things.
- ▶ To enable the communication between the applications and their Things, specialized device drivers need to be initialized on the Fog nodes by the orchestration system.

Network Performance Guarantees

- ▶ Orchestration systems for the Cloud are capable of deploying applications on nodes that can offer the right performance guarantees in terms of processing power, memory, and disk space.
- ▶ Another dimension of complexity arises due to control applications that require network performance guarantees, in terms of upper bounds on latency and jitter, in their communication with Things. In order to support these control applications in the Fog, the orchestration system needs to be able to incorporate network latency and jitter into the application placement and scheduling algorithms.

Data Management

- ▶ **Data in Motion**
- ▶ New data sources are being created and added to the network every day. From a video camera in a transit bus, a tire pressure sensor in a truck, a jet engine, to a smart meter attached to a house.
- ▶ These devices are creating a constant stream of data. Very soon, the data generated by the IoT will make up the majority of all information available on the Internet and will change the face of big data. It will not be possible to store all this data and analyze it later.
- ▶ The real-time nature of these new sources of data requires that their output be evaluated in motion and in meaningful way. **The value of data is often dictated by time—being at its highest value when it is first created.**

Data Management

- ▶ Actionable insights can be extracted and acted upon, as data is generated, to create advantage here and now or even predict the future.
- ▶ Mastery of data—moving from data to wisdom—has the potential to improve various aspects of our personal and business life.
- ▶ Organizations can make better decisions, provide enhanced experiences, and achieve competitive advantage.
- ▶ Most of the new data that will be generated in the IoT is real-time data that fits into a broad category called **Data in Motion**. This refers to the constant stream of sensor-generated data that defies traditional processes for capture, storage, and analysis.

DIKW pyramid

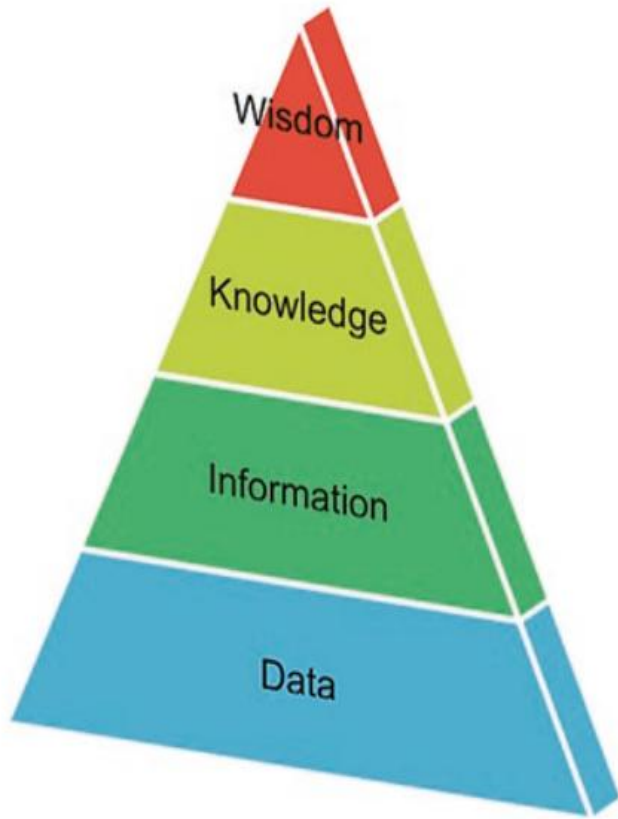


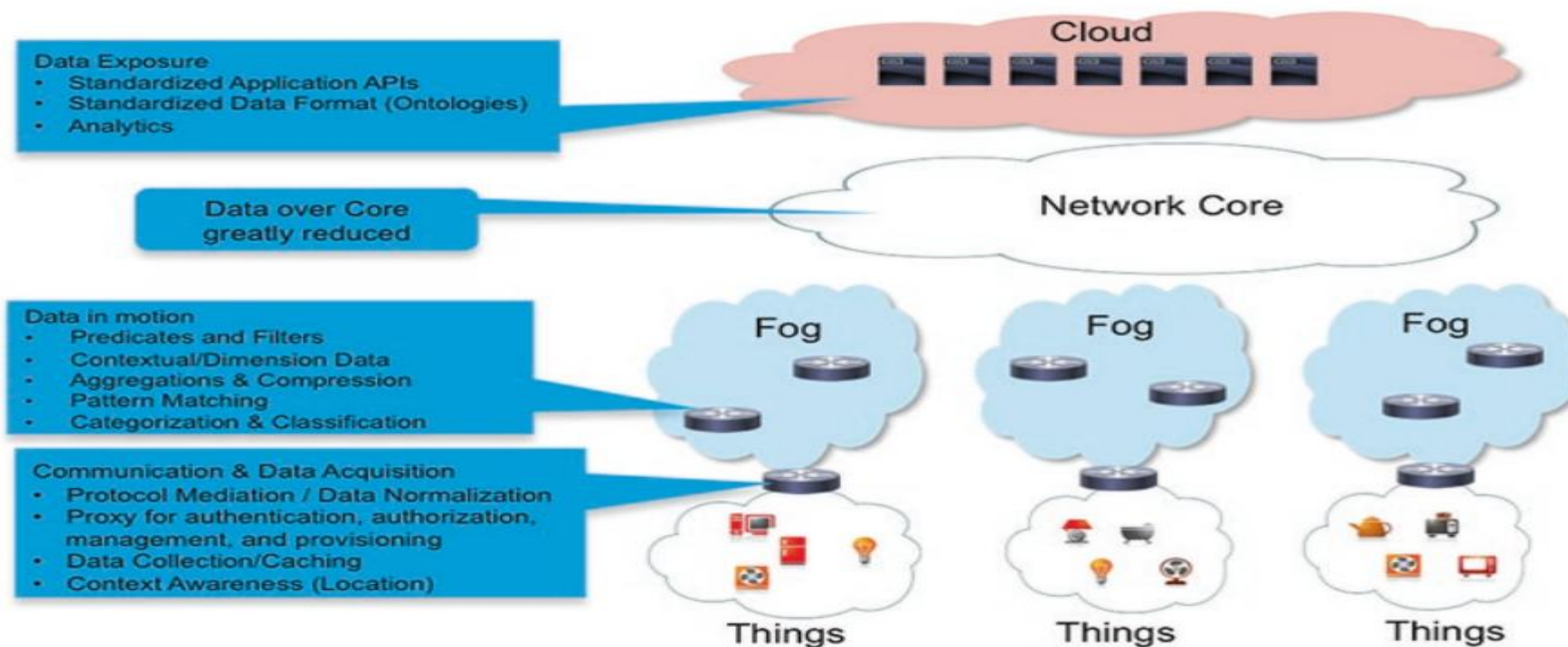
Fig. 6.11 DIKW pyramid

Most of the new data that will be generated in the IoT is real-time data that fits into a broad category called Data in Motion.

Historically, enterprises have focused their analytics or business intelligence applications on data captured and stored using traditional relational data warehouses or “enterprise historian” technologies.

Data management in fog

- Hence, the traditional data management paradigm where raw data is stored first and analyzed later does not fit the temporal nature of IoT data. A new paradigm for handling Data in Motion is required, where data is analyzed as soon as it is generated and then optionally stored if required. The analysis such as: aggregation, reduction/filtering, categorization /classification, contextualization, dimensioning, compression, pattern matching, normalization, and anonymization.



Search Technologies and Engines

- ▶ IoT requires a solution for distributed data search, where queries can be propagated throughout the Fog domains. The solution can be logically organized into two planes: **Things Plane** and **Search Plane**.
- ▶ The Things Plane encompasses the physical Things, Network, and Compute nodes in the Fog.
- ▶ The Search Plane is a logical view of the various Fog nodes that support the distributed search functionality together with the network overlay that enables communication between them.

Search Technologies and Engines

- ▶ Search queries are injected into the Search Plane at some Fog node and propagate throughout the Search Plane.
- ▶ Special considerations are required to ensure that such propagation does not lead to traffic storms that overwhelm the network or the Fog nodes.
- ▶ Wave algorithm and Echo algorithms typically result in very low latency, have a low overhead, and generally scale to hundreds of thousands of nodes. (fig)

Data search in Fog

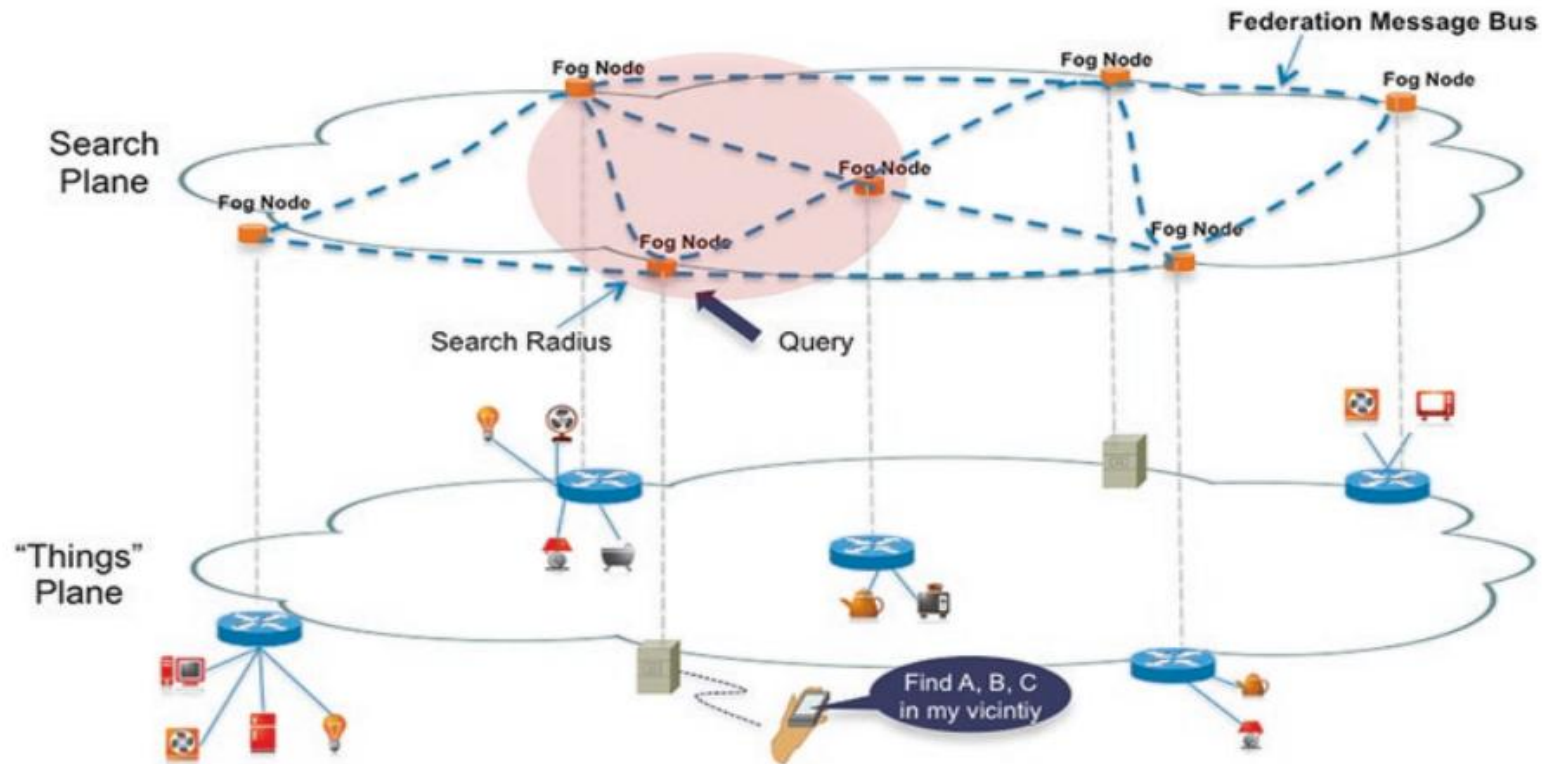


Fig. 6.13 Data search in Fog

Search Technologies and Engines

- ▶ This provides a syntactic search capability with binary (yes/no) outcomes based on exact matches. Exact matches are highly unlikely in real- world IoT deployments with heterogeneous devices and Things from different vendors and providers. E.g. searching “fuzzy matches,” with partial correspondence between the request and the available data.
- ▶ Such mechanisms, ideally, would provide a measure of the semantic similarity between the original request and the retrieved results. To achieve this, Semantic Web technologies could be applied to the IoT:

More Gaps Ahead

- ▶ Clouds are deployed in data centers, where network topologies are well defined and the infrastructure is physically secured with solid walls and cages. Network input and output between applications deployed in the data center and the outside world (e.g., Internet) are mediated through security appliances, such as firewalls, which provide applications with a well-incubated environment under which they can operate.
- ▶ Furthermore, network bandwidth is abundant, and it is relatively easy to change the network physical topology.
- ▶ With Fog, applications may be logically grouped together but not necessarily part of the same physical set up.
- ▶ The first gap to address is providing an orchestration system that enables the connection of applications deployed on Fog nodes to other applications, which are part of the same group, but are on desperate Fog nodes, as well as to applications that are in the Cloud.

More Gaps Ahead

- ▶ These connections could be over bandwidth-constrained links that cannot be changed due to the physical realities of the deployment.
- ▶ Fog nodes may be mounted in the field or on top of a light pole, so anyone could potentially gain physical access to them, attach wires, and compromise the security of the application or the network connectivity.
- ▶ Furthermore, in order for applications to leverage Fog, a high-level programming model is required which simplifies the development of large-scale distributed software.

Summary

- ▶ In this chapter we introduced the concept of Fog computing and discussed its relationship to Cloud computing. The various IoT requirements driving the need for Fog were covered.
- ▶ We also discussed the prerequisites and enabling technologies for Fog, in terms of virtualization technologies, network mobility technologies, orchestration, and data management technologies.

References

- ▶ 1. Book - Internet of Things from hype to reality
the road to digitization (Chapter 6)
By Ammar Rayes and Samer Salam

Practice code

- ▶ Data Reduction: Instead of sending 10 individual HTTP requests to the Cloud, we send zero or one. This significantly reduces network congestion.
- ▶ Latency: The "decision" to alert or discard happens locally. If the `send_to_cloud` function fails due to a lost internet connection, the Fog node could still trigger a local alarm (Edge autonomy).
- ▶ Hierarchical Logic: Edge: The raw `current_temp` generation.
- ▶ Fog: The buffer and average calculation (Processing near the source).
- ▶ Cloud: The final `send_to_cloud` storage/long-term analysis.

Practice code

- ▶ **Scalability:** If you have 1,000 sensors, why is it better to have 10 Fog nodes (each handling 100 sensors) rather than 1,000 sensors talking directly to 1 Cloud server?
- ▶ *Hint: Think about "Single Point of Failure" and "Core Network Traffic."*

Question



Thank you

